AMR and Hospital Length of Stay in India — A Management Case Study

A service improvement perspective developed from Power BI dashboard analysis, linked to the dissertation "Impact of Antimicrobial Resistance (AMR) on Length of Hospital Stays in India: A Scoping Review"

Background

Antimicrobial resistance (AMR) is one of the most serious threats to hospital care in India. When an infection does not respond to standard antibiotics, patients need longer treatment, stronger drugs, and closer monitoring. My scoping review explored how AMR affects the length of hospital stay in Indian hospitals. This project turns that question into an interactive dashboard, to show how a hospital manager could use routine data to understand and act on the problem.

Problem Statement

Across 1,000 admissions in seven hospitals, 62.9% of patients had a resistant infection. Resistance was strongly linked to both longer stays and higher mortality. On average, a resistant infection kept a patient in hospital 9.4 days longer than a non-resistant one.

In short: most infected patients are carrying resistant organisms, and every level of resistance adds more days in hospital and more risk of death.

Key Findings

1. Length of stay rises with every level of resistance. Average stay increased from 5.0 days for sensitive infections, to 10.8 days (MDR), 17.9 days (XDR) and 27.6 days (PDR). A pan-resistant infection keeps a patient in hospital more than five times longer than a treatable one.

2. Mortality rises in the same pattern. The death rate increased from 1.8% for sensitive infections, to 10.0% (MDR), 26.0% (XDR) and 41.5% (PDR).

3. The largest bed burden comes from the middle, not the extreme. PDR patients have the longest individual stays, but they are only 5% of admissions. MDR patients make up 38% of admissions and used the most bed days overall (about 3,800), more than XDR (about 3,100) and more than double PDR (about 1,500).

4. The pattern is consistent across hospitals. Resistance rates ranged from 58% to 68% across the seven hospitals, with excess length of stay between 8.9 and 9.9 days at every site. This points to a system-wide issue rather than a problem at one hospital.

What This Means for Hospital Management

The data suggests two separate management priorities:

• Clinical risk is concentrated in the small group of XDR and PDR patients, who face the longest stays and the highest mortality.

• Operational pressure is driven mainly by the large MDR group. Because there are so many of them, their extra days in hospital add up to the biggest share of bed capacity.

A manager focused only on the worst cases would miss where most of the beds are actually going.

Recommendations

1. Strengthen antimicrobial stewardship. Support a stewardship programme that reviews antibiotic prescribing, so that resistance is slowed at its source rather than only managed after it appears.

2. Prioritise infection prevention and control. Hand hygiene, isolation of resistant cases, and environmental cleaning reduce the spread of resistant organisms between patients, especially in ICUs.

3. Speed up diagnosis. Faster culture and sensitivity results mean patients move onto the right antibiotic sooner, which can shorten stays.

4. Plan bed capacity around the MDR group. Use bed-day data, not only case severity, when planning capacity and discharge support, since MDR patients drive most of the bed pressure.

5. Monitor AMR routinely with a dashboard. Review resistance rate, excess length of stay and mortality monthly, broken down by ward and hospital, so emerging problems are spotted early.

How I Would Measure Success

• Resistance rate: a gradual reduction from 62.9%, as the main long-term indicator of stewardship and infection control impact.

• Excess length of stay: a reduction from 9.4 days, showing that resistant infections are being identified and treated faster.

• Total bed days for MDR patients: tracked monthly, as the clearest measure of operational capacity released.

• Mortality rate by resistance level: monitored to make sure improvements reach the highest-risk XDR and PDR groups.

How the Dashboard Was Built

The analysis was built in Power BI using three linked tables (admissions, hospitals and organisms) arranged as a star schema with a date table. The raw data was cleaned in Power Query, including fixing inconsistent text, standardising Yes/No values, removing duplicate records, handling invalid ages, and calculating length of stay. Nine DAX measures were written to calculate the key indicators, including resistance rate, excess length of stay, mortality rate and total bed days.

Data note: this dashboard uses *synthetic data* generated for learning and portfolio purposes. It is designed to reflect patterns reported in AMR research, but it is not real patient data and the figures should not be cited as evidence.